

1 Method

1.1 Data Collection

1.1.1 Trait selection

We selected traits based on three hypotheses proposed to explain the evolution of masting. For the predator satiation hypothesis, we selected traits associated with seed attractiveness to predators and the potential for surviving in seed caches, including seed weight, fruit size, oil content, dispersal mode and seed dormancy. For the pollination coupling hypothesis, we aimed for traits related to higher pollination efficiency: pollination mode and reproductive type. For resource matching hypothesis, we included traits responsible for higher flexibility in resource use under varying environmental conditions.

1.1.2 Data scraping

For our analyses, we obtained the full species list from the two volumes of *Silvics of North America* Burns & Honkala (1990a,b) (hereafter referred as *Silvics*). We extracted masting data primarily from *Silvics*, using MASTREE as a supplementary source when there was no description on the seed production patterns in *Silvics*. Pollination mode and seed dispersal data were also obtained mainly from *Silvics*. We obtained seed weight data from the Seed Information Database by the Society for Ecological Restoration; gathered seed dormancy data from *Seeds* by Baskin and Baskin and accessed the leaf longevity and oil content data through the TRY database. We compiled fruit size and drought tolerance data from multiple sources, including: *Silvics*, the USDA Fire Effects Information System, the USA National Phenology Network, and The Gymnosperm Database.

We defined masting based on qualitative descriptions of seed production patterns. Although recent studies emphasize quantitative definitions of masting, such data were unavailable from *Silvics*. Instead, we classified species according to key phrases describing their seed production patterns. Species described as producing "a good crop almost every year" were classified as weak masting species, whereas species described as producing "a good crop every several years" or having "irregular seed production" were classified as strong masting species.

We adjusted the data on traits related to masting hypotheses to facilitate easier statistical analysis. For numerical traits with a range (seed weight and fruit size), we recorded the full range from the source, and took the mean value for further analyses. For categorical traits (pollination mode, reproductive type and dispersal mode), we grouped them into broader categories. For example, we classified pollination mode into wind, animal (bird, bat, or insect), or both. The complete raw and processed datasets are available through KNB.

1.2 Statistical Analysis

1.2.1 Masting pattern and single trait

To analyze the relationship between masting pattern and individual traits, we used `phyloglm` function in the `phylolm` package (v. 2.6.2), treating masting as a binary response variable. To construct the phylogenetic tree, we verified all species names using the `WorldFlora` package (v. 1.14-5) and updated all species names according to WCVB Backbone data (2025). We then constructed an ultrametric phylogenetic tree in R using the `phytools` (v. 2.4-4) and `ape` (v. 5.7-1) packages. We calculated the half-life using the equation $(\log(2)/\alpha)$ and compared it with the tree height (standardized to 1). This metric represents the time required for the influence of ancestral states to decrease by half.

1.2.2 Masting pattern with a multivariate analysis

To conduct a multivariate analysis for our categorical traits, we selected non-metric multidimensional scaling (NMDS) with the `metaMDS` function in the `vegan` (v. 2.7-1) package and evaluated both two- and three-dimensional solutions $k = 2, 3$ and selected $k = 3$ based on the lower stress score. To check for potential patterns, we plotted trait distributions on two panels (NMDS1 vs. NMDS2 and NMDS2 vs. NMDS3).

1.2.3 Phylogenetic signal for individual traits

To check for the phylogenetic signal for our traits, we used different metrics for different trait types. For binary traits, we calculated Fritz and Purvis' D-statistics using the function `phylo.d` in the `caper` (v. 1.0.4) package. For continuous traits, we calculated Pagel's λ using the function `phylosig` in the `phytools` (v. 2.4-4) package. For categorical traits, we estimated the Pagel's λ using the function `fitDiscrete` in the `geiger` (v. 2.0.11) package.

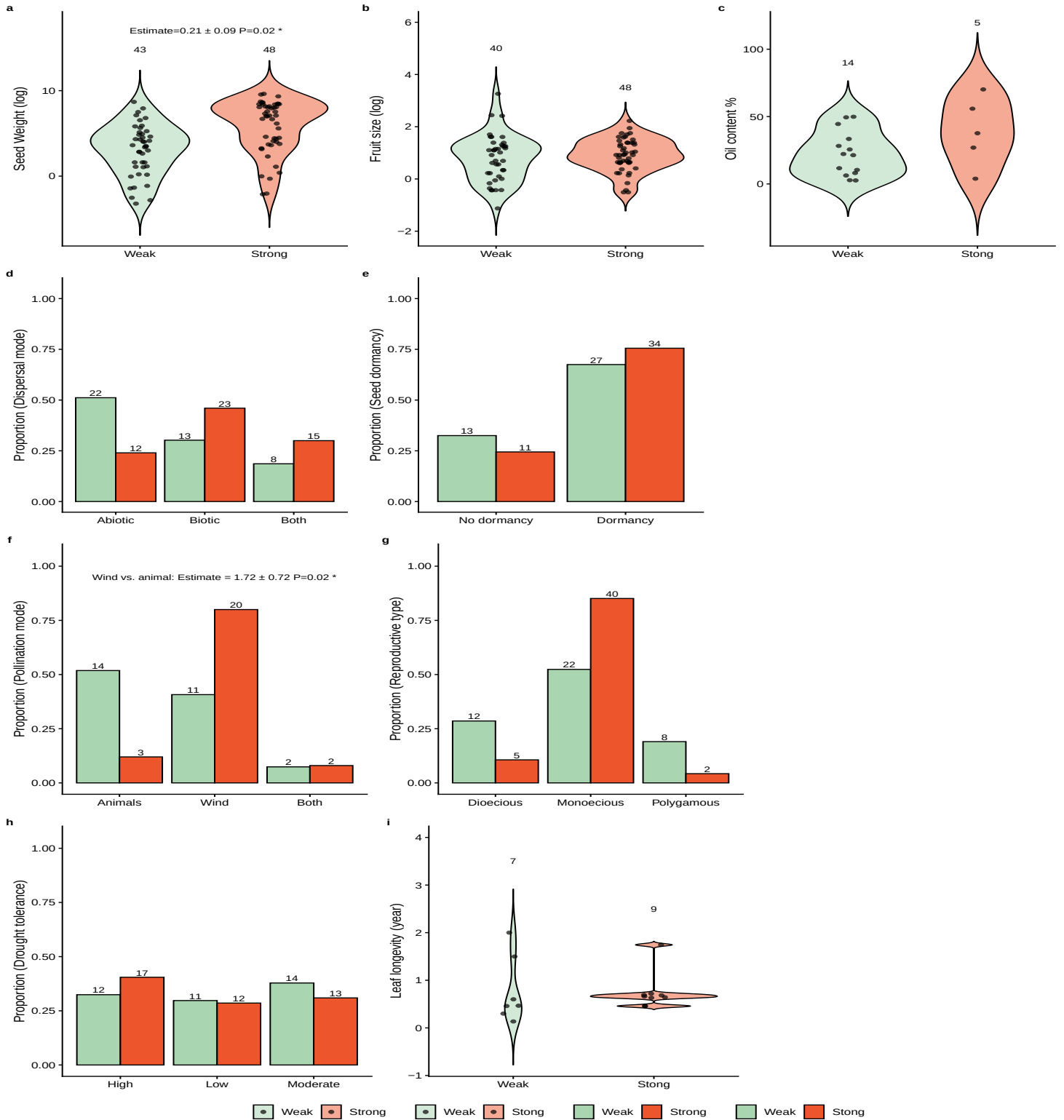


Figure 1: How plant traits related to different hypotheses differ in strong masting and weak masting for angiosperm species: We show data for traits related to the predator satiation hypothesis (a-e), pollination coupling hypothesis (f-g) and the resource matching hypothesis (h-i) as either violin plots (a-c, i) or proportional histograms (d-h) separated by species categorized as strong masting (red) or weak masting (green). We show model summaries only where significant (P -value < 0.5) and also report the model sample size (N), regression coefficient (Estimate), and phylogenetic signal parameter α . Full model results are show in Table S1.

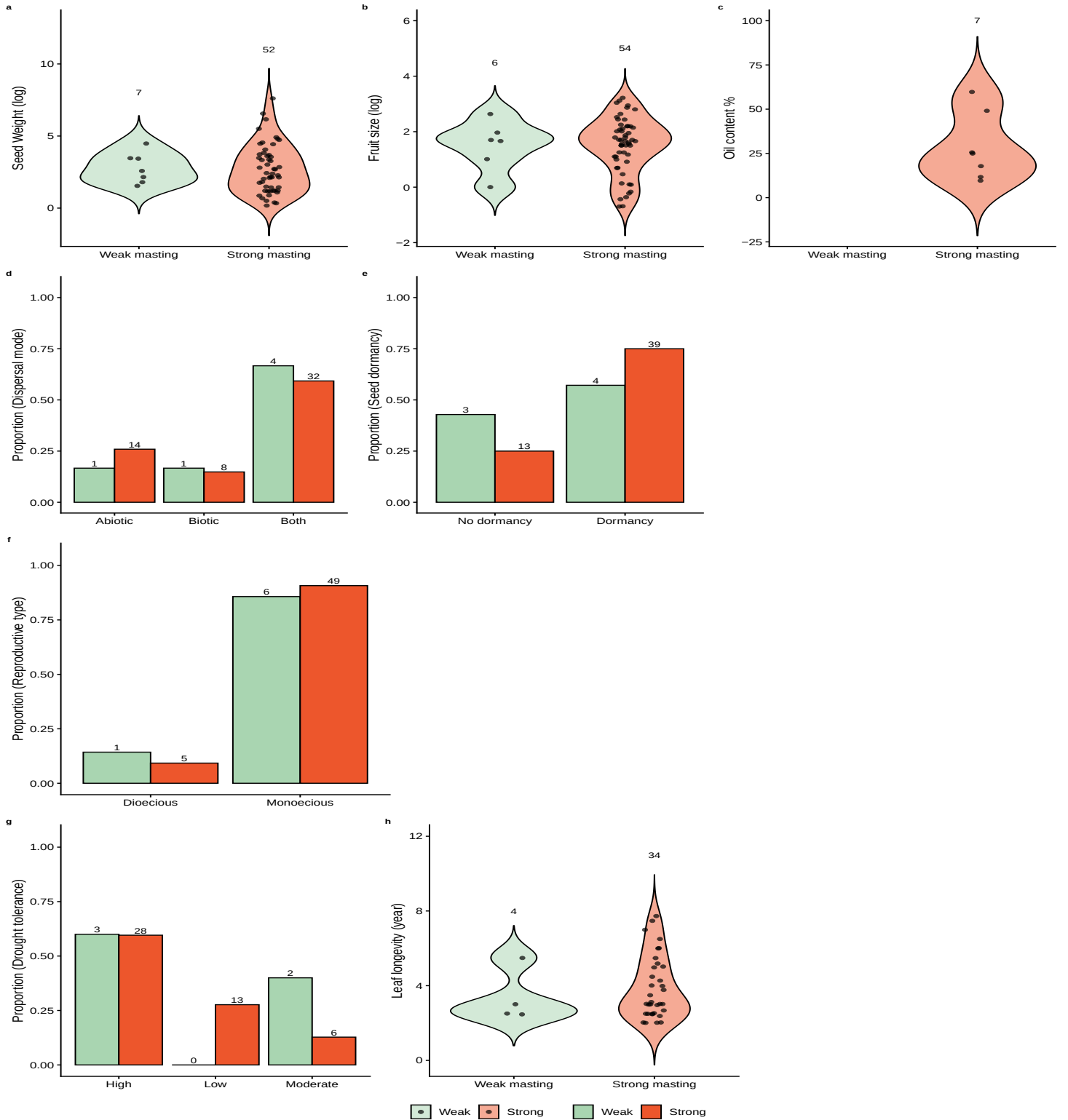


Figure 2: How plant traits related to different hypotheses differ in strong masting and weak masting for gymnosperm species: We show data for traits related to the predator satiation hypothesis (a-e), pollination coupling hypothesis (f-g) and the resource matching hypothesis (h-i) as either violin plots (a-c, i) or proportional histograms (d-h) separated by species categorized as strong masting (red) or weak masting (green). We show model summaries only where significant (P -value < 0.5) and also report the model sample size (N), regression coefficient (Estimate), and phylogenetic signal parameter α . Full model results are show in Table S2.

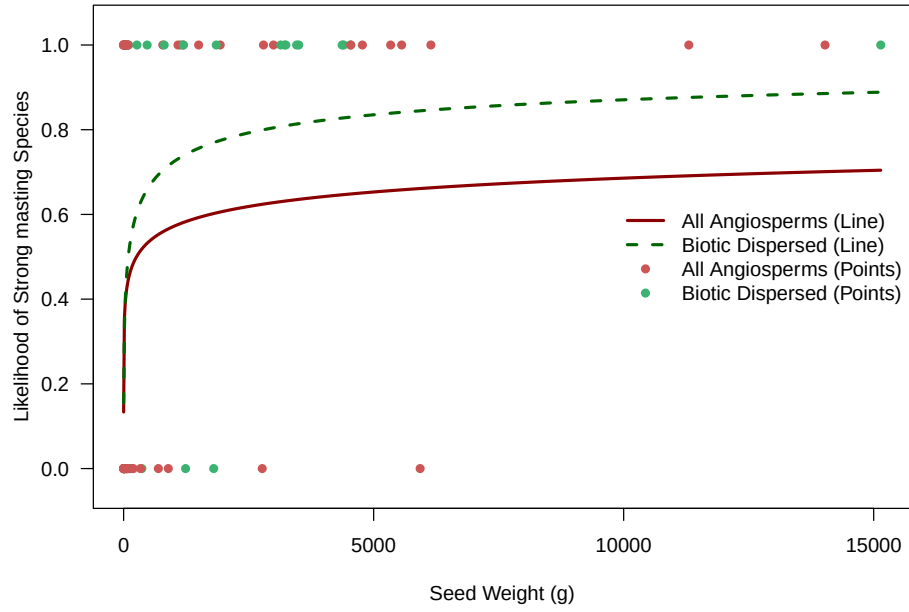


Figure 3: Predicted likelihood of masting as a function of seed weight for all angiosperm species (solid red line) and the biotic-dispersed subgroup (dashed green line), estimated using phylogenetic logistic regression. Individual species observations are plotted as binary data points at 0 (weak masting) and 1 (strong masting).

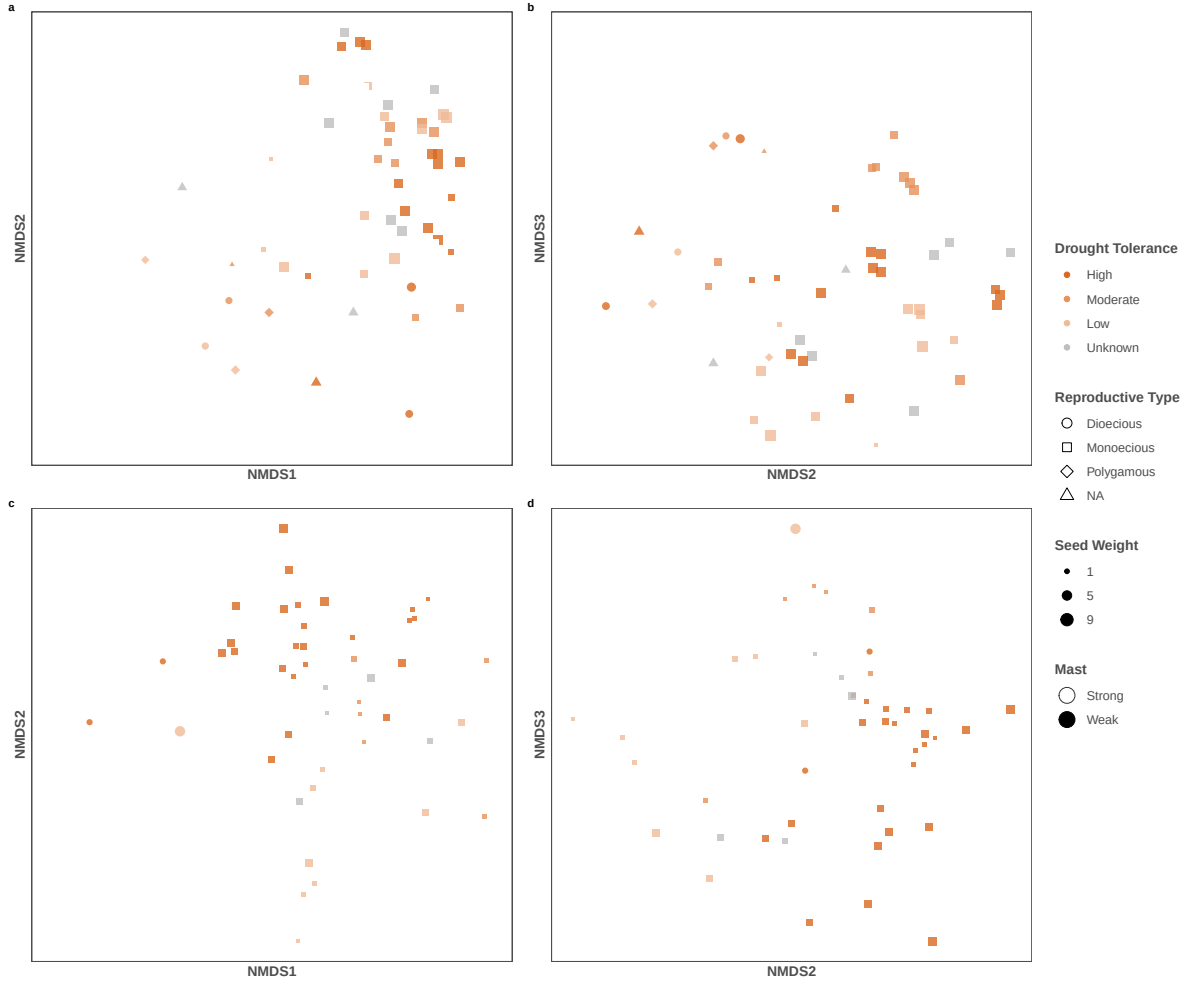


Figure 5: Non-metric multidimensional scaling (NMDS) of trait composition for angiosperm and gymnosperm species. Each point represents a species, with position determined by NMDS axes. Points are colored by drought tolerance, filled by strong masting or not, shaped by reproductive type, sized by seed weight. Panels are arranged as (a) angiosperm NMDS1–NMDS2, (b) angiosperm NMDS2–NMDS3, (c) gymnosperm NMDS1–NMDS2, and (d) gymnosperm NMDS2–NMDS3.

2 Results

2.0.1 Structure of Data

In total, our dataset included 188 species, comprising 124 angiosperm species and 64 gymnosperm species. Among angiosperms, 79% (98/124) of species had masting data available, with strong masting and weak masting species occurring in approximately equal proportions 52% (51/98). Among gymnosperms, 95% (61/64) of species had masting data available, and nearly all were classified as strong masting species 89% (54/61). We were not able to get complete trait data for every single trait for all species with masting data. The traits with the most available data was seed weight. The traits with the sparse data were oil content and leaf longevity (Fig. 1 and Fig. 2).

2.0.2 Results for Each Hypothesis

We found relationships between traits related to predator satiation and pollination efficiency and strong masting for angiosperm species. Among the traits related to predator satiation, seed weight was the only traits significantly related to masting (p -value = 0.02, Table S1). Larger seed weight was associated with a greater probability of strong masting (Figure 1). For angiosperms only, increasing seed weight threefold (from a median seed weight of 90.9g to 272.7g) increased the predicted likelihood of strong masting from 0.44 to 0.5 (Figure 3). The effect was stronger among species with biotically dispersed seeds, where increasing seed weight threefold (from a median seed weight of 1087.15g to 3261.45g) increased the predicted likelihood of strong masting from 0.73 to 0.81 (Table S3; Figure 3).

Among traits related to pollination efficiency, pollination mode was the only pollination-related trait significantly associated with strong masting (p -value = 0.02, Table S1) for angiosperm. Wind-pollinated species had a predicted likelihood of strong masting of 0.59, compared with 0.2 for animal-pollinated species. All gymnosperm species in our dataset were wind-pollinated, and gymnosperms contained a much higher proportion of strong masting species overall. This can also indicate that wind pollination is associated with strong masting.

2.0.3 Multivariate analysis

Strong and weak masting species were not clearly separated along any axis in the NMDS analysis (Fig. 5). We found no evidence that combinations of traits associated with any of the three hypotheses explained variation in masting pattern.

2.0.4 Phylogenetic signal

Masting pattern showed a moderate phylogenetic signal in both gymnosperms ($D = 0.51$) and angiosperms ($D = 0.58$) (Table S5).

Seed weight has strong phylogenetic signal for both angiosperm ($\lambda = 0.96$) and gymnosperm ($\lambda = 0.98$). This pattern is also evident in Fig. 4. Dispersal mode, leaf longevity, pollination mode and reproductive type also showed high phylogenetic signal in Table S4. However, these values should be interpreted cautiously because Pagel's λ is estimated differently for categorical and continuous traits, making direct comparisons inappropriate.

3 Discussion

We found that masting was associated with traits related to both pollination and seed dispersal as well as establishment, providing support for multiple mechanisms underlying interannual variation in seed production. The positive relationship between wind pollination and strong masting in angiosperm supports the pollination coupling hypothesis, while the positive relationship between seed weight and strong masting provides evidence consistent with the predator satiation hypothesis. These associations point to selective pressure acting at both early and late stages of the reproductive cycle, which are critical for successful reproduction and gene transmission.

Previous literature have also reported positive relationship between seed mass and masting, with one explanation being that bigger seeds require more resource investment and simply cannot reproduce every year. However, from an evolutionary perspective, the association between seed mass and masting may reflect more than the resource costs of producing large seeds. For predator satiation to favour synchronized reproduction, seed crops must provide a sufficiently large resource to overwhelm seed consumers. For some species, seed predators are also seed dispersers. Larger seeds probably make them more attractive to these mammals and help with the further seed dispersal. At the same time, larger seeds generally contain greater nutritional reserves, which can enhance seedling establishment and early survival following dispersal. Thus, the positive association between seed weight and masting may reflect coordinated reproductive strategies in which synchronized production reduces the relative impact of seed predation while larger seeds increase the probability of successful establishment. Together, these processes could generate selection favouring reproductive strategies that enhance seed dispersal and survival.

3.1 Scale dependence in trait–masting relationships

Recent work has emphasized that masting is better represented as a continuous rather than categorical trait, which is mostly available at the population level. The coefficient of variation (CV) is one of the most widely used continuous measures of masting. However, the use of CV as a measure of masting has been widely discussed, as it does not capture all characteristics of masting and can be strongly distorted by years with zero seed production. At the same time, evaluating the evolutionary associations between traits and masting requires comparisons across species. If a trait has a sufficiently strong and consistent association with the evolution of masting, we expect this relationship to be detectable at the species level using binary masting data.

At the same time, many of the traits considered in our analysis are phenotypically plastic and may vary among populations, potentially in response to local environmental conditions. Thus, traits for which we fail to detect a species-level association with masting may still be important at the population level. There are certain species only exhibit masting under certain environmental conditions. For example, *Hymenaea coubaril* has been reported to mast only on the island where seed predators are present, suggesting that the expression of masting can depend on local ecological conditions. As a result, understanding how reproductive traits are associated with masting may require measurements of both traits and reproductive variation at the population level, considering local environmental conditions and intraspecific variation.

3.2 Distinguishing selection across types of masting

Masting has originally been studied primarily as interannual variation in seed production, but recent work has refined the definition between seed masting and flower masting. These two forms of reproductive synchrony occur at different stages of the reproductive cycle: seed masting focuses on the final reproductive outcome, whereas flower masting occurs at the beginning of reproduction. Importantly, variation in flowering and seed production does not necessarily occur at the same time, suggesting that these two forms of masting may be shaped by different selective pressures. For example, we would expect mechanisms related to pollination to perform stronger selection on flowering synchrony, whereas mechanisms related to seed predation, dispersal and establishment may be more directly associated with synchrony in seed production. Separating species according to their dominant form of masting could therefore provide a more refined test of hypotheses underlying reproductive synchrony. Examining the associations between different types of masting and corresponding subset of reproductive traits therefore is necessary for us to understand how different selective pressures contribute to the evolution of reproductive synchrony at different stages of the reproductive cycle.

3.3 Phylogenetic signal limits inference

We found strong phylogenetic signal for many traits we selected and also masting has a phylogenetic signal, so for the analysis we did, it is hard to tease out effect of phylogeny. Also, many traits we included have relatively sparse data, and these data are nested in certain clade. This makes the phylogenetic bias a potentially bigger problem.

3.4 Deep evolutionary split between angiosperm and gymnosperm

It is also important to look for the evidence of evolution separately for angiosperm and gymnosperm.

3.5 Importance of testing within clade

References

- Burns, R.M. & Honkala, B.H. (eds.) (1990a) *Silvics of North America*, vol. 1 of *Agriculture Handbook*. United States Department of Agriculture, Forest Service, Washington, DC.
- Burns, R.M. & Honkala, B.H. (eds.) (1990b) *Silvics of North America*, vol. 2 of *Agriculture Handbook*. United States Department of Agriculture, Forest Service, Washington, DC.

Cooper, N., Thomas, G.H., Venditti, C., Meade, A. & Freckleton, R.P. (2016) A cautionary note on the use of ornstein uhlenbeck models in macroevolutionary studies. *Biological journal of the Linnean Society* **118**, 64–77.

A Supplementary Material

Trait	Type	Predictor	N	Estimate	SE	P	Half-life
Seed weight (log)	Numeric	Seed weight (log)	91	0.21	0.09	0.02	0.08
Fruit size (log)	Numeric	Fruit size (log)	88	0.08	0.31	0.79	0.04
Oil content (%)	Numeric	Oil content (%)	19	0.04	0.03	0.20	0.04
Seed dormancy	Categorical	Dormant vs. non-dormant	85	0.05	0.51	0.92	0.06
Dispersal mode	Categorical	Biotic vs. abiotic	93	0.53	0.53	0.32	0.09
Dispersal mode	Categorical	Both vs. abiotic	93	0.53	0.57	0.35	0.09
Pollination mode	Categorical	Wind vs. animal	52	1.72	0.72	0.02	0.01
Pollination mode	Categorical	Wind vs. animal and animals	52	1.18	1.15	0.31	0.01
Reproductive type	Categorical	Monoecious vs. dioecious	89	0.76	0.68	0.26	0.05
Reproductive type	Categorical	Monoecious vs. polygamous	89	-0.29	0.96	0.76	0.05
Drought tolerance	Categorical	Low vs. high drought tolerance	79	0.15	0.47	0.76	0.06
Drought tolerance	Categorical	Moderate vs. high drought tolerance	79	-0.15	0.45	0.74	0.06
Leaf longevity (years)	Numeric	Leaf longevity (years)	16	0.26	0.95	0.78	0.77

Table S1: Phylogenetic generalized linear model results for angiosperm species. We modeled each trait separately with masting as a binary variable and traits as either categorical or continuous (Type column). For categorical traits, we report the the difference in log-odds between a given level and its designated baseline reference level; for continuous traits, we report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor) and the half-life. The half-life represents the time required for the influence of ancestral states to decrease by half; where larger values suggests the residuals are more affected by phylogenetic history than the environment (and vice versa for smaller values). Given out trees are scaled to 1, a value near 1 thus means phylogenetic correlation persists across the phylogenetic depth. For more interpretation, see Cooper *et al.* (2016).

Trait	Type	Predictor	N	Estimate	SE	P	Half-life
Seed weight (log)	Numeric	Seed weight (log)	59	0.01	0.24	0.98	0.01
Fruit size (log)	Numeric	Fruit size (log)	60	0.03	0.36	0.94	0.01
Seed dispersal	Categorical	Biotic vs. abiotic	60	-0.83	0.81	0.30	0.36
Seed dispersal	Categorical	Both vs. abiotic	60	-1.11	0.67	0.10	0.36
Seed dormancy	Categorical	Dormant vs. non-dormant	59	0.69	0.69	0.32	0.01
Reproductive type	Categorical	Monoecious vs. dioecious	61	-1.13	1.01	0.26	0.78
Drought tolerance	Categorical	Low vs. high drought tolerance	52	11.36	98.16	0.91	34.66
Drought tolerance	Categorical	Moderate vs. high drought tolerance	52	-0.29	0.60	0.63	34.66
Leaf longevity (years)	Numeric	Leaf longevity (years)	38	0.15	0.35	0.67	0.01

Table S2: Phylogenetic generalized linear model results for gymnosperm species. We modeled each trait separately with masting as a binary variable and traits as either categorical or continuous (Type column). For categorical traits, we report the the difference in log-odds between a given level and its designated baseline reference level; for continuous traits, we report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor) and the half-life. The half-life represents the time required for the influence of ancestral states to decrease by half; where larger values suggests the residuals are more affected by phylogenetic history than the environment (and vice versa for smaller values). Given out trees are scaled to 1, a value near 1 thus means phylogenetic correlation persists across the phylogenetic depth. For more interpretation, see Cooper *et al.* (2016).

Trait	Group	Predictor	N	Estimate	SE	P	Half-life
Seed weight (log)	Biotic and both	Seed weight (log)	55	0.41	0.16	0.01	0.01
Seed weight (log)	Abiotic	Seed weight (log)	36	0.12	0.13	0.36	0.01

Table S3: Phylogenetic generalized linear model results for only seed weight for biotic-dispersed subgroup (both group included) and abiotic-dispersed subgroup. We report the change in log-odds per unit increase of the predictor (Estimate column). We also report the standard error (SE), P-value (with $P < 0.05$ considered significant and marked with an asterisk next to the Predictor) and the half-life.

Group	Trait	λ
Angiosperm	Seed weight (Log)	0.95
Angiosperm	Fruit size (Log)	0.93
Angiosperm	Oil content	0.25
Angiosperm	Leaf longevity	1.02
Angiosperm	Dispersal mode	1.00
Angiosperm	Pollination mode	1.00
Angiosperm	Reproductive type	0.00
Angiosperm	Drought tolerance	0.00
Gymnosperm	Seed weight (Log)	0.98
Gymnosperm	Fruit size (Log)	0.91
Gymnosperm	Oil content %	0.00
Gymnosperm	Leaf longevity	0.90
Gymnosperm	Dispersal mode	0.00
Gymnosperm	Reproductive type	1.00
Gymnosperm	Drought tolerance	0.00

Table S4: Phylogenetic Signal (Pagel's λ) for continous traits and multi-level categorical traits. $\lambda = 0$ indicates no phylogenetic signal, trait districuted independent of the phylogeny; $0 < \lambda < 1$ means moderate phylogenetic signal, trait is partially structured by the phylogeny; $\lambda = 1$ indicates strong phylogenetic signal, trait follows Brownian motion.

Group	Trait	Estimated D	P(random)	P(Brownian)
Angiosperm	Masting	0.58	0	0.01
Angiosperm	Seed dormancy	0.25	0	0.16
Gymnosperm	Masting	0.51	0.02	0.10
Gymnosperm	Seed dormancy	0.98	0.41	0

Table S5: Phylogenetic Signal (D-statistic) for the phylogenetic signal of binary traits, with probabilities calculated under a random phylogenetic structure and under a Brownian motion phylogenetic structure. $D = 1$ indicates the trait is randomly distributed across the phylogeny; $D = 0$ indicates the trait follows a Brownian motion model; $0 < D < 1$ indicates moderate phylogenetic signal; $D < 0$ indicates the trait is more clumped than expected under Brownian motion; and $D > 1$ indicates overdispersion. P random is the p-value for the null hypothesis that the trait is randomly distributed across the phylogeny (no phylogenetic signal), and P Brownian is the p-value for the null hypothesis that the trait follows a Brownian motion model (strong phylogenetic signal). A low p-value indicates rejection of the corresponding null hypothesis.